INTERNATIONAL MONETARY FUND

SENEGAL

SEVENTH REVIEW UNDER THE POLICY SUPPORT INSTRUMENT AND REQUEST FOR MODIFICATION OF ASSESSMENT CRITERIA—DEBT SUSTAINABILITY ANALYSIS

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Senegal remains at a low risk of debt distress. Under the baseline scenario, which is consistent with higher program ceilings for nonconcessional and semi-concessional borrowing, all the debt burden indicators remain way below their policy-dependent indicative thresholds, and debt ratios in present value terms are lower than in the previous debt sustainability analysis (DSA) reflecting the use of a higher discount rate. There are in stress tests small and temporary breaches of the threshold on the debt to GDP ratio and on the debt service to revenue ratio, the latter corresponding to the repayment of two Eurobonds. In addition, the probability approach shows a favorable outlook, reflecting Senegal's high CPIA score.[1] The DSA, however, suggests that there is not much space for higher fiscal deficits if the low risk rating is to be preserved. It also indicates a need for caution in the recourse to nonconcessional borrowing.[2]

[1] Senegal's CPIA score was 3.82 in 2012, and on average 3.76 over 2010–12. Under the debt sustainability framework rules, this corresponds to a "moderate" performance. Maintenance of an average CPIA score above 3.75 would ultimately lead to a reclassification to "high" performance.

[2] The DSA presented in this document is based on the LIC DSF Guidance note (2013).

BACKGROUND

1. Senegal's total public debt and external debt ratios have both increased substantially over the past five years. The ratio of total public debt to GDP amounted to about 47 percent in 2013, up from around 25 percent in 2008. At the same time, the stock of total external public and publicly guaranteed debt has increased from around 20 percent of GDP in 2008 to around 32 percent at end-2013. These levels are close to those that prevailed before Senegal benefited from debt relief under the Multilateral Debt Relief Initiative (MDRI) in 2006. Debt-servicing costs have also increased, reflecting higher recourse to market financing. External public debt service has increased from 4.3 percent of exports in 2008 to about 9 percent in 2013.

Total External Debt, Central Government

	(Percent of GDP, as of end of year)						% of total
	2008	**2009**	**2010**	**2011**	**2012**	**2013**	**2013**
Total	19.6	26.8	27.3	27.4	31.0	32.4	100.0
Multilateral creditors	12.0	17.5	18.1	16.5	19.0	20.7	63.7
IDA/IBRD	7.0	8.0	8.0	8.8	9.2	9.6	29.6
AfDB/AfDF	1.5	2.2	2.5	2.9	3.0	3.6	11.0
IMF	0.5	1.8	2.9	0.0	1.9	2.4	7.4
OFID/BADEA/IsDB	1.7	3.2	2.7	2.8	2.6	2.6	8.0
EIB	0.2	0.3	0.3	0.3	0.2	0.3	0.8
Others	1.2	2.0	1.8	1.7	2.1	2.2	6.8
Bilateral creditors	7.6	7.9	7.6	7.2	8.1	0.1	26.2
OECD countries	2.2	2.7	2.6	2.1	3.2	3.3	10.1
Arab countries	4.1	3.8	2.8	2.2	2.1	2.0	6.1
Others	1.2	1.3	2.1	2.3	2.8	2.8	8.8
Commercial creditors	0.0	1.5	1.6	3.7	3.5	3.3	10.1
Memorandum Item							
Nominal GDP (CFAF billions)	5994	6050	6395	6775	7165	7308	

Sources: Senegalese authorities and IMF staff estimates

2. In terms of composition, the bulk of Senegal's debt remains largely external, and provided on concessional terms, but the use of financial market instruments is increasing.[3] Most of the public debt is external (i.e., owed to non-residents of the West African Economic and Monetary Union, WAEMU), although the share of domestic debt to GDP has increased from 5.3 percent in 2008 to around 16 percent in 2013.[4] Almost two thirds of Senegal's external debt is owed to multilateral creditors—primarily the World

[3] Public debt refers to the debt of the central government.

[4] Domestic debt includes debt issued in the WAEMU financial market.

Bank and the African Development Bank. The largest groups of bilateral creditors are the OECD and the Arab countries. The share of market debt is still relatively small, although it has grown rapidly over the past few years, with the issuance of two Eurobonds in 2009 and 2011, and the contracting of a syndicated loan

in 2013, which includes a tranche in euro targeting nonresidents. This is expected to be followed by a US$500 million Eurobond in mid-2014. Private external debt has averaged about 20 percent of GDP over the last decade and was estimated at about 36 percent of GDP at end-2013.[5] About half of this debt was in the form of trade credits and bank deposits; the rest consisted of debt securities, loans, and other liabilities. This exposure was partly offset by private external assets amounting to 8 percent of GDP.

BORROWING PLAN AND UNDERLYING ASSUMPTIONS

3. The authorities have continued to strengthen their capacity to manage debt and assess project loans. Senegal has continued to record improvement in its sub-CPIA score, showing marked progress in debt management. A new public debt directorate has been created, combining units that previously managed domestic debt and external debt separately. In addition, Senegal's first medium-term debt strategy was completed in the fall of 2012 and a second one is being finalized. Progress is underway to improve project appraisal and selection, in particular by developing capacity to conduct cost-benefit analysis. In light of these favorable developments, Senegal was upgraded to the "higher capacity" category during the 2013 assessment of macroeconomic and public financial management capacity (see Classification of Low-Income Countries for the Purpose of Debt Limits in Fund-Supported Programs).

4. This DSA is consistent with the macroeconomic framework outlined in the staff report. As in the June 2013 DSA, the baseline scenario assumes the implementation of sound macroeconomic and structural policies, leading to an increase in economic growth and a narrowing of fiscal deficits over the long term. Other notable features include:

- **Real GDP growth** is expected to increase to slightly above 5 percent, as in the previous DSA. This assumes efficiency gains from reform implementation, which would allow total factor productivity (TFP) to resume growing. TFP declined in recent years, which affected growth performance.

- **Fiscal deficit.** The overall fiscal deficit projections are broadly unchanged beyond 2014 and in line with the authorities' commitment to meet the key WAEMU convergence criterion on the fiscal deficit.[6]

[5] Estimates of private sector external debt are based on BCEAO data on the international investment position.

[6] The convergence criterion calls for the basic fiscal balance, in percent of GDP, to be greater than or equal to zero. The basic fiscal balance is defined as total revenue, plus budget support grants, plus the counterpart of HIPC/MDRI-related spending, minus current expenditures and domestically-financed capital expenditure.

- **Current account deficit**: The current account deficit is projected to narrow gradually from 10.8 percent in 2013 to 8 percent of GDP in 2018 and slightly further down in the long term. This would be driven by projected fiscal consolidation and by the strong dynamism of exports (mining in particular). Remittances are projected to remain significant as a share of GDP.

- **Inflation:** it is expected to remain moderate, on average close to 2.0 percent.

- **Financing:** The authorities are increasingly relying on external nonconcessional or semi-concessional borrowing to finance infrastructure projects, and this trend is expected to continue. They postponed a planned Eurobond issuance to 2014, following a sharp tightening of financial conditions on international markets in the course of 2013. The authorities now intend to issue a US$500 million Eurobond in mid-2014. Conditions have been relatively favorable in international markets in the past few months, and the authorities expect to get a much lower interest rate than on their 2011 Eurobond (whose yield is currently about 6 percent). Part of the proceeds would be used to repay the euro tranche of the syndicated loan contracted in 2013, which has a short maturity. The projections assume a rollover of the 2011 and 2014 bonds at maturity, as well as a moderate annual amount nonconcessional borrowing in the medium and long term. The authorities intend to continue relying on semi-concessional project financing (i.e. with a grant element above 15 percent) for infrastructure. As a result, the average grant element of new external borrowing is projected to decrease from about 20 percent to below 10 percent over the projection period, as Senegal gradually moves away from concessional borrowing toward nonconcessional borrowing.

- **Discount rate:** A discount rate of 5 percent has been used for this DSA.[7]

Evolution of Selected Macroeconomic Indicators

	2012	2013	2014	2015	Long term 1/
Real GDP growth					
Current DSA	3.4	3.5	4.9	5.0	5.4
Previous DSA	3.5	4.0	4.6	4.7	5.3
Overall fiscal deficit (percent of GDP)					
Current DSA	5.9	5.5	5.1	3.9	2.6
Previous DSA	5.9	5.3	4.6	3.9	2.8
Current account deficit (percent of GDP)					
Current DSA	10.8	10.4	10.0	8.9	7.6
Previous DSA	10.3	9.3	8.7	8.3	7.6

1/ Defined as the last 15 years of the projection period. For the current DSA, the long term covers the years 2020-2034; for the previous DSA, it covers 2019-2033.

[7] In line with the IMF Executive Board decision to reform the system of discount rates used in the analysis of debt in low-income countries, a single uniform discount rate of 5 percent is used to calculate the present value (PV) of external debt and in calculating the grant element of loans under the program. The discount rate used in the 2013 DSA was 3 percent.

EXTERNAL DSA

5. In Senegal, remittances are both high as a percentage of GDP and stable. Since 2000, they have grown every year with the exception of 2009, when they fell 6 percent. Over the period 2010–2012, remittances in Senegal averaged 52 percent of exports of goods and services and 13 percent of GDP. They have become an important and reliable source of foreign exchange in Senegal, a pattern that is expected to continue.[8]

6. Taking remittances into account, debt burden indicators remain well below their thresholds in the baseline scenario. The external PPG debt ratios remain below their respective thresholds even under the most extreme stress tests, with two exceptions. Two spikes in debt service reflect the assumption of the repayment of the Eurobonds at maturity, and lead to one-year breaches under the most extreme stress test (a 30 percent depreciation of the currency). There is also a small temporary breach of the threshold on the debt to GDP ratio under the most extreme stress test. These breaches disappear using the probability approach, which incorporates more country specific features for Senegal.[9]

PUBLIC DSA

7. Under the PSI baseline scenario, indicators of overall public debt (external plus domestic) do not show significant vulnerabilities. The PV of total public debt to GDP and the PV of total public debt to revenue remain gradually decrease over the projection period. The PV of public debt to GDP remains well below the benchmark level of 56 percent associated with heightened public debt vulnerabilities for medium performers. The authorities' effort to increase maturities (from slightly over one year at the time of the previous DSA) should reduce exposure to rollover and interest rate risks in the context of financing from the regional market.

8. The public debt outlook would be much less favorable in the absence of fiscal consolidation (Table 2b). In a scenario that assumes an unchanged primary deficit (as a percent of GDP) over the entire projection period, the PV of public debt to GDP grows rapidly, quickly breaching the 56 percent benchmark level. The benchmark level is also breached in the "historical" scenario (holding real GDP growth and the primary deficit constant at their historical levels). While overall the risks remain low, these stress tests highlight the importance of continuing the fiscal effort and raising output growth.

OTHER ISSUES

9. Views of the authorities and assessment of Plan Sénégal Emergent (PSE). The authorities broadly agreed with the staff's assessment of debt sustainability using the PSI baseline scenario. They

[8] Both ratios are measured on a backward-looking, three-year average basis.

[9] According to the debt sustainability framework policy, the probability approach can be used in borderline cases, when breaches are short-lived and within plus-or-minus 10 percent of the indicative threshold.

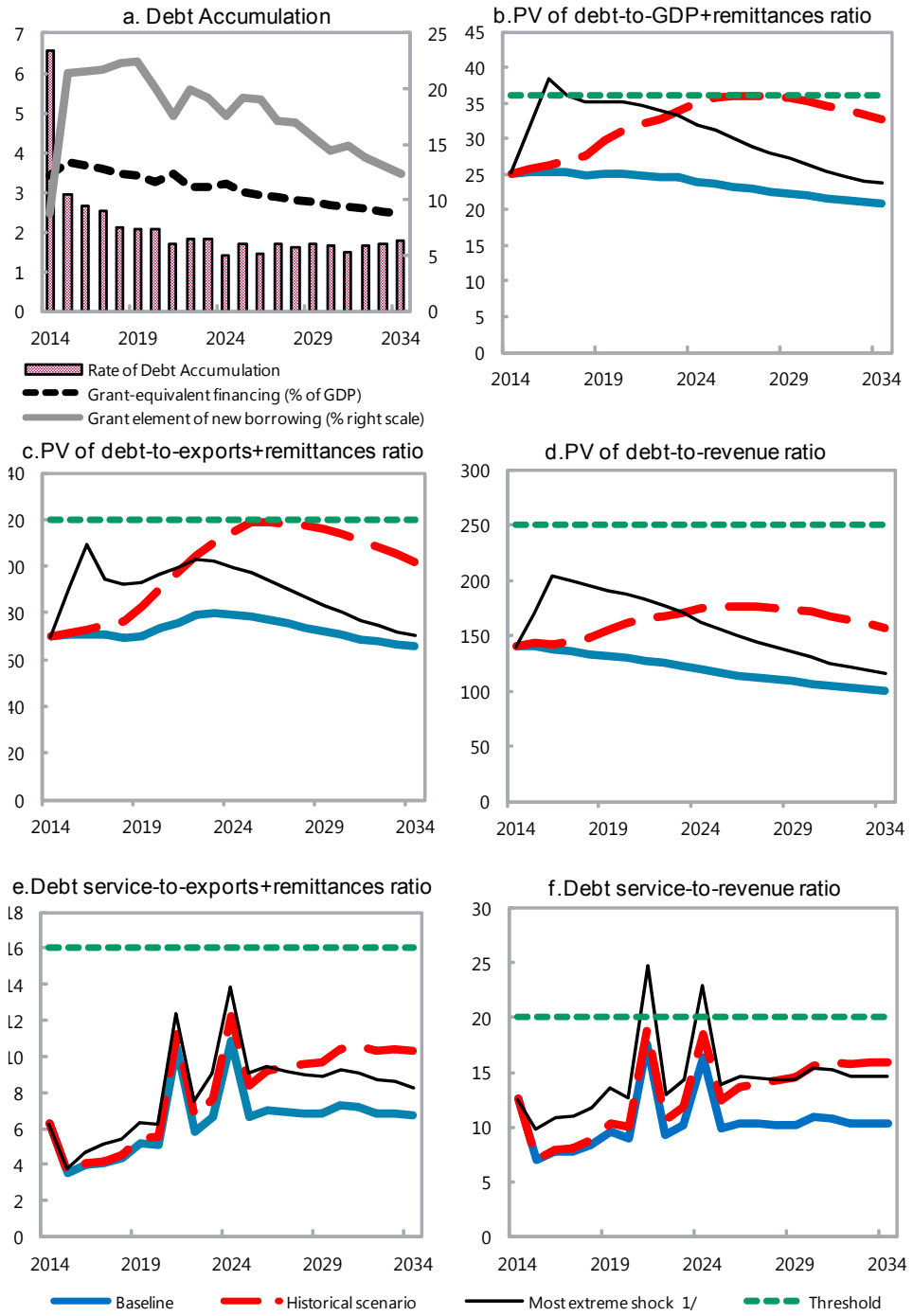
highlighted their recent fiscal consolidation efforts, pointing to actions taken to reduce the fiscal deficit in 2012–13 and further in 2014. Most of the discussion focused on the implications of the PSE for debt dynamics. Under the macroeconomic framework developed by the authorities for the PSE, the expected growth dividend from implementing the PSE is assumed to eventually offset the impact on debt ratios of higher fiscal deficits in the next few years (Figure 3). The authorities nevertheless acknowledged that fiscal consolidation and a cautious approach to nonconcessional borrowing were critical for safeguarding debt sustainability. Staff stressed that preserving debt sustainability under the PSE would depend on achieving a high growth dividend and implementing a comprehensive and ambitious reform of the state (to make room for investment and improve the efficiency of spending). In the absence of balance of payments projections consistent with the authorities' PSE framework, the external debt impact of the PSE is not considered in this DSA.

10. Revised debt limits. The authorities and staff discussed the options for debt limits made available by Senegal's recent reclassification as a "higher capacity" country under the debt limits policy. In light of the limited time remaining until the end of the current Fund-supported program, it was agreed to keep the current design for debt limits, while raising the existing ceilings as needed, in particular the one on semi-concessional borrowing to facilitate the financing of four high-return infrastructure projects (electricity and sanitation). These higher ceilings are taken into account in the present DSA.

CONCLUSION

11. In staff's view, Senegal continues to face a low risk of debt distress. This assessment, however, hinges critically on a continued reduction of the fiscal deficit and a prudent shift towards less concessional financing.

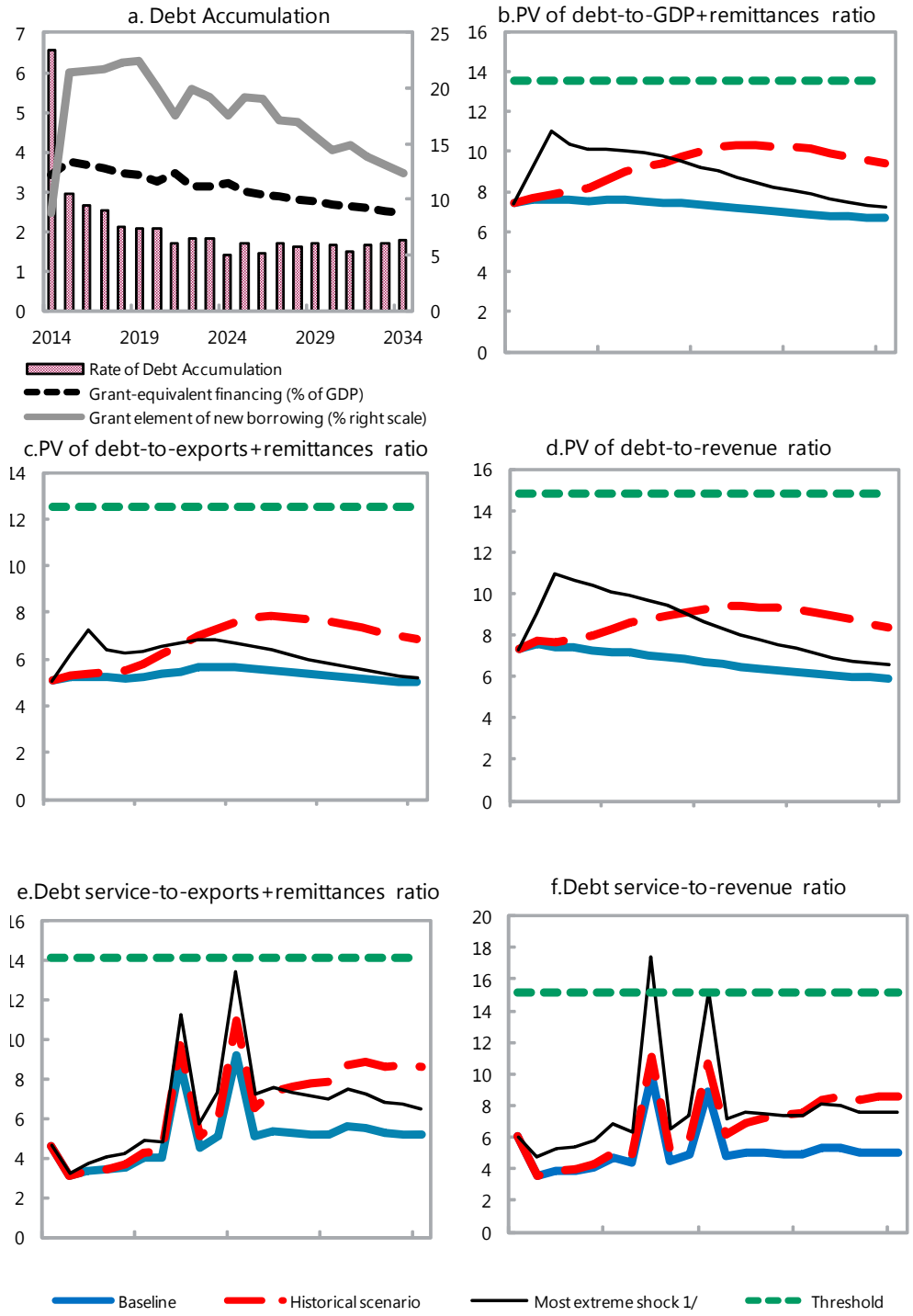
Figure 1. Senegal: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2014-2034 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a Combination shock; in c. to a Combination shock; in d. to a Combination shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

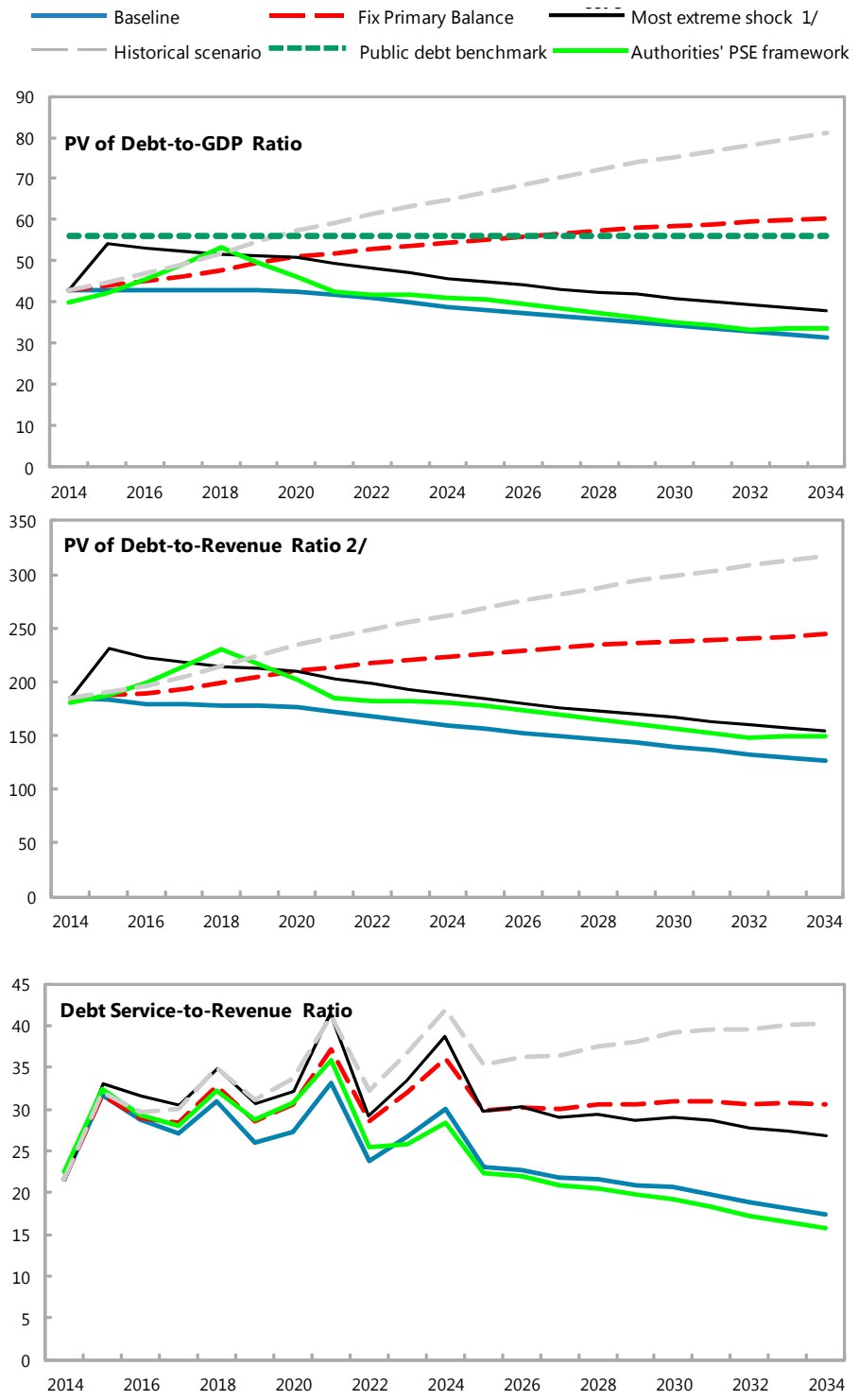
Figure 2. Senegal: Probability of Debt Distress of Public and Publicly Guarantees External Debt under Alternative Scenarios, 2014-2034 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a Combination shock; in c. to a Combination shock; in d. to a Combination shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 3. Senegal: Indicators of Public Debt under Alternative Scenarios, 2014-2034



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Authorities' estimates before revisions to 2014 fiscal indicators based on PSI discussions.

Table 1. Senegal: External Debt Sustainability Framework, Baseline Scenario, 2011-2034 1/

(In percent of GDP, unless otherwise indicated

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2014-2019 Average	2024	2034	2020-2034 Average
External debt (nominal) 1/	**54.5**	**62.7**	**68.3**			**73.0**	**75.5**	**77.3**	**78.3**	**77.9**	**76.8**		**69.7**	**73.2**	
of which: public and publicly guaranteed (PPG)	29.4	31.0	32.4			35.6	36.1	36.3	36.3	35.8	35.6		32.6	27.4	
Change in external debt	3.8	8.1	5.7			4.7	2.5	1.9	0.9	-0.3	-1.1		-2.7	0.8	
Identified net debt-creating flows	0.8	10.1	5.5			5.2	3.7	3.3	2.8	2.2	0.9		1.3	2.8	
Non-interest current account deficit	**6.9**	**9.9**	**9.6**	**8.2**	**2.9**	**9.0**	**7.9**	**7.7**	**7.4**	**7.0**	**5.8**		**6.0**	**7.6**	6.9
Deficit in balance of goods and services	18.3	21.3	21.8			21.3	20.6	20.5	20.2	19.6	17.8		14.7	15.9	
Exports	26.4	28.3	27.5			27.3	26.8	26.7	26.6	26.9	27.3		23.8	25.8	
Imports	44.7	49.5	49.3			48.6	47.3	47.1	46.8	46.6	45.0		38.5	41.8	
Net current transfers (negative = inflow)	-12.3	-12.5	-12.6	-10.9	2.1	-12.4	-12.8	-12.8	-12.7	-12.6	-11.9		-8.7	-8.3	-8.7
of which: official	-0.9	-1.0	-0.3			-0.5	-0.6	-0.6	-0.5	-0.5	-0.5		-0.5	-0.5	
Other current account flows (negative = net inflow)	0.9	1.2	0.3			0.1	0.1	0.0	0.0	0.0	-0.1		-0.1	0.0	
Net FDI (negative = inflow)	**-2.0**	**-2.0**	**-1.7**	**-1.8**	**0.6**	**-1.7**	**-1.8**	**-1.9**	**-2.0**	**-2.0**	**-2.1**		**-2.1**	**-2.1**	-2.1
Endogenous debt dynamics 2/	**-4.1**	**2.2**	**-2.4**			**-2.1**	**-2.3**	**-2.5**	**-2.6**	**-2.7**	**-2.8**		**-2.7**	**-2.7**	
Contribution from nominal interest rate	1.0	0.9	0.8			1.0	1.0	1.0	1.0	1.0	1.0		0.9	0.9	
Contribution from real GDP growth	-0.8	-1.9	-2.1			-3.1	-3.3	-3.5	-3.6	-3.7	-3.8		-3.6	-3.7	
Contribution from price and exchange rate changes	-4.3	3.2	-1.1			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**3.0**	**-2.0**	**0.2**			**-0.5**	**-1.3**	**-1.4**	**-1.9**	**-2.6**	**-2.0**		**-3.9**	**-2.0**	-2.4
of which: exceptional financing	-0.9	-1.4	-0.1			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	60.2			66.1	68.2	69.9	70.7	70.4	69.5		63.4	68.7	
In percent of exports	...	...	219.0			242.4	255.0	262.3	265.4	261.3	255.1		266.3	265.8	
PV of PPG external debt	**...**	**...**	**24.3**			**28.6**	**28.9**	**28.9**	**28.8**	**28.3**	**28.3**		**26.2**	**22.8**	
In percent of exports	**...**	**...**	**88.3**			**104.9**	**107.9**	**108.5**	**107.9**	**105.1**	**103.9**		**110.2**	**88.3**	
In percent of government revenues	**...**	**...**	**120.6**			**140.2**	**140.2**	**137.2**	**135.7**	**132.9**	**131.4**		**119.3**	**100.4**	
Debt service-to-exports ratio (in percent)	**17.8**	**9.7**	**8.9**			**11.6**	**7.1**	**7.3**	**7.1**	**7.4**	**8.4**		**15.7**	**8.0**	
PPG debt service-to-exports ratio (in percent)	**13.2**	**7.0**	**6.7**			**9.4**	**5.4**	**6.1**	**6.2**	**6.6**	**7.6**		**15.1**	**9.1**	
PPG debt service-to-revenue ratio (in percent)	**17.2**	**9.6**	**9.2**			**12.6**	**7.0**	**7.7**	**7.8**	**8.3**	**9.7**		**16.3**	**10.4**	
Total gross financing need (Billions of U.S. dollars)	1.4	1.5	1.5			1.7	1.4	1.5	1.5	1.6	1.5		2.8	6.1	
Non-interest current account deficit that stabilizes debt ratio	3.1	1.8	3.9			4.3	5.5	5.9	6.5	7.3	6.9		8.7	6.8	
Key macroeconomic assumptions															
Real GDP growth (in percent)	1.7	3.4	3.5	3.8	1.4	4.9	5.0	5.1	5.1	5.1	5.3	5.1	5.4	5.5	5.4
GDP deflator in US dollar terms (change in percent)	9.3	-5.5	1.8	4.3	8.0	4.6	4.1	3.5	3.9	3.8	2.3	3.7	2.6	2.7	2.6
Effective interest rate (percent) 5/	2.2	1.6	1.4	1.8	0.4	1.5	1.5	1.4	1.4	1.4	1.4	1.4	1.3	1.4	1.4
Growth of exports of G&S (US dollar terms, in percent)	17.7	4.7	2.5	8.8	10.7	8.8	7.3	8.4	9.2	10.3	9.1	8.8	8.9	9.5	7.8
Growth of imports of G&S (US dollar terms, in percent)	23.2	8.2	4.9	11.9	17.2	8.1	6.5	8.4	8.5	8.6	4.2	7.4	6.2	9.3	7.6
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	8.8	21.5	21.6	21.7	22.3	22.5	19.7	17.6	12.3	16.7
Government revenues (excluding grants, in percent of GDP)	20.3	20.4	20.1			20.4	20.6	21.1	21.2	21.3	21.6		22.0	22.7	22.2
Aid flows (in Billions of US dollars) 7/	0.7	1.0	0.8			0.9	1.1	1.1	1.2	1.3	1.4		1.8	3.1	
of which: Grants	0.3	0.4	0.4			0.5	0.5	0.5	0.6	0.6	0.7		0.9	1.7	
of which: Concessional loans	0.4	0.6	0.4			0.5	0.6	0.6	0.6	0.7	0.7		0.9	1.5	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			3.4	3.7	3.7	3.6	3.5	3.4		3.2	2.5	2.9
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			34.8	53.8	54.5	54.9	56.7	57.2		46.9	45.8	50.7
Memorandum items:															
Nominal GDP (Billions of US dollars)	14.4	14.0	14.8			16.2	17.7	19.3	21.1	23.0	24.8		36.5	80.4	
Nominal dollar GDP growth	11.1	-2.3	5.4			9.8	9.3	8.8	9.3	9.1	7.8	9.0	8.1	8.3	8.1
PV of PPG external debt (in Billions of US dollars)			3.7			4.7	5.2	5.6	6.1	6.6	7.0		9.6	18.3	
(PVt-PVt-1)/GDPt-1 (in percent)						**6.6**	2.9	2.7	2.5	2.1	2.0	3.1	1.4	1.8	1.7
Gross workers' remittances (Billions of US dollars)	1.9	1.9	2.1			2.2	2.5	2.7	3.0	3.2	3.3		3.4	7.2	
PV of PPG external debt (in percent of GDP + remittances)	...	...	21.3			25.1	25.3	25.3	25.2	24.8	25.0		24.0	20.9	
PV of PPG external debt (in percent of exports + remittances)	...	...	58.2			69.7	70.6	70.9	70.6	69.1	70.1		79.0	65.4	
Debt service of PPG external debt (in percent of exports + remittance	...	...	4.4			6.2	3.5	4.0	4.1	4.3	5.2		10.8	6.8	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Senegal: Public Sector Sustainability Framework, Baseline Scenario, 2011-2034

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate						Projections			
	2011	2012	2013	Average 5/	Standard Deviation 5/	2014	2015	2016	2017	2018	2019	2014-19 Average	2024	2034	2020-34 Average
Public sector debt 1/	40.7	43.4	46.8			50.0	50.3	50.4	50.4	50.4	50.3		45.4	36.1	
of which: foreign-currency denominated	29.4	31.0	32.4			35.6	36.1	36.3	36.3	35.8	35.6		32.6	27.4	
Change in public sector debt	5.2	2.7	3.4			3.2	0.3	0.1	0.0	0.0	0.0		-1.1	-0.7	
Identified debt-creating flows	4.9	3.2	3.3			2.0	-0.1	-0.3	-0.7	-0.8	-0.5		-1.2	-0.7	
Primary deficit	5.2	4.4	3.9	3.9	1.1	3.3	2.3	1.9	1.7	1.5	1.1	2.0	0.8	0.7	0.8
Revenue and grants	22.5	23.3	22.7			23.2	23.4	23.9	24.0	24.0	24.2		24.4	24.8	
of which: grants	2.2	2.9	2.6			2.8	2.9	2.8	2.8	2.7	2.7		2.4	2.1	
Primary (noninterest) expenditure	27.7	27.7	26.6			26.5	25.7	25.8	25.6	25.5	25.4		25.2	25.5	
Automatic debt dynamics	-0.3	-0.6	-0.6			-1.3	-2.4	-2.3	-2.4	-2.3	-1.7		-1.9	-1.5	
Contribution from interest rate/growth differential	-0.4	-0.7	0.7			-1.1	-1.8	-1.8	-1.9	-1.9	-1.9		-1.9	-1.5	
of which: contribution from average real interest rate	0.2	0.6	2.2			1.1	0.5	0.6	0.6	0.6	0.6		0.4	0.5	
of which: contribution from real GDP growth	-0.6	-1.3	-1.5			-2.2	-2.4	-2.4	-2.5	-2.5	-2.5		-2.4	-1.9	
Contribution from real exchange rate depreciation	0.1	0.1	-1.3			-0.2	-0.5	-0.5	-0.5	-0.5	0.2		...	...	
Other identified debt-creating flows	0.0	-0.6	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	-0.6	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	0.3	-0.5	0.1			1.2	0.4	0.5	0.7	0.8	0.5		0.1	0.0	
Of which: asset changes	1.9	1.6	0.1			0.3	0.3	0.3	0.3	0.3	0.2		0.0	0.0	
Other Sustainability Indicators															
PV of public sector debt	...	...	38.7			43.0	43.0	43.0	42.9	42.8	43.0		39.0	31.5	
of which: foreign-currency denominated	...	...	24.3			28.6	28.9	28.9	28.8	28.3	28.3		26.2	22.8	
of which: external	...	...	24.3			28.6	28.9	28.9	28.8	28.3	28.3		26.2	22.8	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	12.3	12.2	12.2			14.6	13.0	11.0	10.3	10.9	9.2		9.3	5.6	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	170.5			185.5	183.5	179.9	179.1	178.4	177.6		159.6	127.2	
PV of public sector debt-to-revenue ratio (in percent)	...	...	192.2			211.1	209.0	203.8	202.4	201.2	199.6		177.3	138.7	
of which: external 3/	...	...	120.6			140.2	140.2	137.2	135.7	132.9	131.4		119.3	100.4	
Debt service-to-revenue and grants ratio (in percent) 4/	19.9	16.4	17.0			21.6	31.7	28.7	27.1	30.9	25.9		30.1	17.4	
Debt service-to-revenue ratio (in percent) 4/	22.1	18.7	19.2			24.6	36.1	32.5	30.7	34.8	29.2		33.4	19.0	
Primary deficit that stabilizes the debt-to-GDP ratio	0.0	1.7	0.5			0.1	2.0	1.8	1.7	1.6	1.2		1.8	1.4	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	1.7	3.4	3.5	3.8	1.4	4.9	5.0	5.1	5.1	5.1	5.3	5.1	5.4	5.5	5.4
Average nominal interest rate on forex debt (in percent)	3.7	2.5	2.5	2.4	0.6	2.9	2.8	2.8	2.8	2.8	2.8	2.8	2.8	3.6	3.0
Average real interest rate on domestic debt (in percent)	3.7	4.8	8.1	3.2	3.7	4.9	2.7	3.3	3.3	3.2	3.4	3.5	3.1	2.7	3.0
Real exchange rate depreciation (in percent, + indicates depreciation	0.3	0.5	-4.3	-0.8	8.5	-0.7	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	4.2	2.3	-1.4	2.4	2.8	1.5	2.3	2.3	2.3	2.3	2.3	2.2	2.6	2.7	2.6
Growth of real primary spending (deflated by GDP deflator, in percer	7.9	3.2	-0.4	1.1	2.6	4.4	1.9	5.5	4.3	4.8	4.6	4.3	5.0	5.6	5.4
Grant element of new external borrowing (in percent)	...	...	...	...	...	8.8	21.5	21.6	21.7	22.3	22.5	19.7	17.6	12.3	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 3A. Senegal: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014-2034 (In Percent)

	Projections							
	2014	2015	2016	2017	2018	2019	**2024**	2034
PV of debt-to-GDP+remittances ratio								
Baseline	25	25	25	25	25	25	**24**	21
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	25	26	26	27	28	30	**35**	33
A2. New public sector loans on less favorable terms in 2014-2034 2	25	27	27	28	28	28	**30**	31
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	25	26	27	27	26	26	**25**	22
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	25	27	31	30	30	30	**27**	21
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	25	27	29	29	28	29	**27**	24
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	25	30	34	32	31	31	**28**	22
B5. Combination of B1-B4 using one-half standard deviation shocks	25	31	38	36	35	35	**32**	24
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	25	34	34	34	33	34	**33**	28
PV of debt-to-exports+remittances ratio								
Baseline	70	71	71	71	69	70	**79**	65
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	70	72	73	75	76	82	**115**	102
A2. New public sector loans on less favorable terms in 2014-2034 2	70	74	76	77	77	79	**99**	96
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	70	71	71	71	70	70	**79**	65
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	70	80	96	95	93	94	**102**	77
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	70	71	71	71	70	70	**79**	65
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	70	90	101	89	87	87	**93**	67
B5. Combination of B1-B4 using one-half standard deviation shocks	70	90	109	95	92	93	**99**	70
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	70	71	71	71	70	70	**79**	65
PV of debt-to-revenue ratio								
Baseline	140	140	137	136	133	131	**119**	100
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	140	143	142	145	148	156	**174**	157
A2. New public sector loans on less favorable terms in 2014-2034 2	140	147	147	149	149	149	**149**	147
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	140	145	145	144	141	138	**126**	106
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	140	150	165	162	158	155	**136**	103
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	140	152	160	159	155	153	**138**	117
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	140	160	175	172	167	163	**141**	104
B5. Combination of B1-B4 using one-half standard deviation shocks	140	170	204	200	194	190	**162**	116
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	140	198	194	192	188	185	**168**	141

Table 3B. Senegal: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014-2034 (continued) (In Percent)

Debt service-to-exports+remittances ratio								
Baseline	6	4	4	4	4	5	**11**	7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	6	4	4	4	5	5	**12**	10
A2. New public sector loans on less favorable terms in 2014-2034 2	6	4	3	3	4	5	**8**	9
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	6	4	4	4	4	5	**11**	7
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	6	4	5	5	5	6	**14**	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	6	4	4	4	4	5	**11**	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	6	4	5	5	5	6	**13**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	6	4	5	5	5	6	**13**	8
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	6	4	4	4	4	5	**11**	7
Debt service-to-revenue ratio								
Baseline	13	7	8	8	8	10	**16**	10
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	13	7	8	8	9	10	**18**	16
A2. New public sector loans on less favorable terms in 2014-2034 2	13	7	7	7	7	9	**11**	13
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	13	7	8	8	9	10	**17**	11
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	13	7	8	9	9	11	**18**	11
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	13	8	9	9	10	11	**19**	12
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	13	7	8	9	10	11	**19**	11
B5. Combination of B1-B4 using one-half standard deviation shocks	13	7	9	10	11	12	**22**	13
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	13	10	11	11	12	14	**23**	15
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	16	16	16	16	16	16	**16**	16

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assu
an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 4. Senegal: Sensitivity Analysis for Key Indicators of Public Debt, 2014-2034

	Projections							
	2014	2015	2016	2017	2018	2019	2024	2034
PV of Debt-to-GDP Ratio								
Baseline	43	43	43	43	43	43	39	31
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	43	45	47	49	52	55	65	81
A2. Primary balance is unchanged from 2014	43	44	45	46	48	50	54	60
A3. Permanently lower GDP growth 1/	43	43	43	44	44	44	43	43
Authorities' PSE framework 3/	40	42	46	49	53	50	41	34
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	43	45	47	48	48	49	49	47
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	43	45	48	48	47	47	43	34
B3. Combination of B1-B2 using one half standard deviation shocks	43	46	49	49	50	50	49	44
B4. One-time 30 percent real depreciation in 2015	43	54	53	52	52	51	46	38
B5. 10 percent of GDP increase in other debt-creating flows in 2015	43	52	52	51	51	50	45	35
PV of Debt-to-Revenue Ratio 2/								
Baseline	186	183	180	179	178	178	160	127
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	186	191	196	205	214	225	262	318
A2. Primary balance is unchanged from 2014	186	187	189	194	199	205	223	244
A3. Permanently lower GDP growth 1/	186	184	181	182	182	183	175	174
Authorities' PSE framework 3/	180	187	199	214	231	216	180	150
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	186	190	194	197	200	203	200	190
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	186	194	201	199	197	196	174	136
B3. Combination of B1-B2 using one half standard deviation shocks	186	194	203	204	205	207	198	179
B4. One-time 30 percent real depreciation in 2015	186	231	223	218	215	212	188	154
B5. 10 percent of GDP increase in other debt-creating flows in 2015	186	222	216	213	210	208	185	143
Debt Service-to-Revenue Ratio 2/								
Baseline	22	32	29	27	31	26	30	17
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	22	32	30	30	35	31	42	40
A2. Primary balance is unchanged from 2014	22	32	29	28	33	29	36	31
A3. Permanently lower GDP growth 1/	22	32	29	27	31	27	32	22
Authorities' PSE framework 3/	23	32	29	28	32	29	28	16
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	22	32	30	29	34	29	35	25
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	22	32	29	30	35	27	32	19
B3. Combination of B1-B2 using one half standard deviation shocks	22	32	30	31	35	29	34	24
B4. One-time 30 percent real depreciation in 2015	22	33	32	30	35	31	39	27
B5. 10 percent of GDP increase in other debt-creating flows in 2015	22	32	30	38	33	29	33	20

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.
3/ Authorities' estimates before revisions to 2014 fiscal indicators based on PSI discussions.